



JABchem



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Past Papers Advanced Higher Chemistry

2025 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/160)	%	
A	108+	68%	30.0%
B	92+	58%	25.0%
C	76+	48%	22.2%
D	60+	38%	13.8%
No award	<60	<38%	9.0%

Section:	Multiple Choice	Extended Answer	Project
Average Mark:	17.5 /25	49.1 /95	27.1 /40

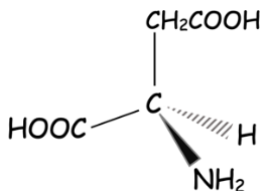
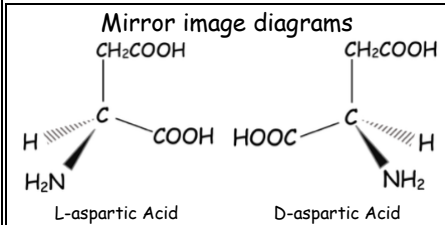
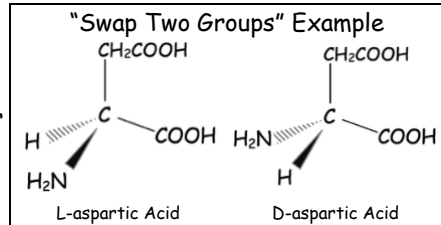
2025 Adv Higher Chemistry Marking Scheme

MC Qu	Answer	% Correct	Reasoning															
1	A	93	☒ A each line corresponds to the exact amount of energy released when an electron drops down from a higher energy level to a lower energy level. ☒ B visible part of electromagnetic spectrum is from 380nm to 750nm ☒ C energy is absorber not emitted when electron moves to higher energy level ☒ D wavelengths emitted are from energy released when electrons drop down energy levels															
2	D	67	☒ A all p-orbitals have a <i>figure of 8</i> shape but no ring in middle as in the diagram ☒ B all p-orbitals have a <i>figure of 8</i> shape but no ring in middle as in the diagram ☒ C only d-orbital d_{z^2} has the shape shown, the other four have double <i>figure of 8</i> shapes ☒ D the d_{z^2} orbital has the shape shown in the diagram															
3	D	61	X^{3+} ion has 55 electrons $\therefore$ an atom of X must have 58 electrons and therefore 58 protons Element with atomic number of 58 is Cerium Ce and is one of the Lanthanides in f-block															
4	B	53	Electron Pairs = $\frac{\text{No. of electrons in outer shell of central atom} + \text{No. of bonds} - \text{Charge}}{2}$ Electron Pairs = $\frac{6 + 4 - 0}{2}$ Electron Pairs = $\frac{10}{2}$ = 5 pairs = Trigonal bipyramidal electron pair arrangement															
5	A	42	<table><tr><td>Complex</td><td>$[\text{Cu}(\text{OH}_2)_6]^{2+}$</td><td>$[\text{CuCl}_4]^{2-}$</td></tr><tr><td>Light Transmitted</td><td>blue</td><td>green</td></tr><tr><td>Light Absorbed</td><td>yellow</td><td>purple</td></tr><tr><td>Wavelengths Absorbed</td><td>590nm-595nm</td><td>380nm-400nm</td></tr><tr><td>Energy of light absorbed</td><td>Higher wavelength = Lower energy</td><td>Lower wavelength = higher energy</td></tr></table> ☒ A increase in energy of light absorbed and increase in splitting of d-orbitals as a result ☒ B increases the splitting of d-orbitals increases the energy of the light absorbed ☒ C energy of light absorbed increases ☒ D energy of light absorbed increases	Complex	$[\text{Cu}(\text{OH}_2)_6]^{2+}$	$[\text{CuCl}_4]^{2-}$	Light Transmitted	blue	green	Light Absorbed	yellow	purple	Wavelengths Absorbed	590nm-595nm	380nm-400nm	Energy of light absorbed	Higher wavelength = Lower energy	Lower wavelength = higher energy
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6	C	39	☒ A all 5 d-orbitals are only degenerate (equal in energy) when there are no ligands present ☒ B the arrangement is the effect of octahedral complex not a tetrahedral arrangement ☒ C in octahedral arrangement d_{xy} , d_{xz} and d_{yz} are lower in energy as they lie along the x,y and z axes while the $d_{x^2-y^2}$ and d_{z^2} orbitals are higher in energy because there are not aligned with the x, y and z axes and repelled more by the ligands In the tetrahedral arrangement the situation is reversed. The d_{xy} , d_{xz} and d_{yz} do not lie along the axes of the tetrahedral arrangement are they are pushed higher in energy by repulsion from ligands and the $d_{x^2-y^2}$ and d_{z^2} orbitals are repelled by the ligands less ☒ D the arrangement is named after legendary Italian international fullback Fuctivano															
7	A	48	☒ A Elemental sulphur has oxidation number zero and S in $\text{S}_2\text{O}_3^{2-}$ ion is +2 ☒ B if $3\text{O}=-6$ and charge of ion is -2 then $2\text{S}=+4 \therefore$ S in $\text{S}_2\text{O}_3^{2-}$ has oxidation number +2 ☒ C S_8 is the element sulphur. All elements have an oxidation number of zero ☒ D S_8 is the element sulphur. All elements have an oxidation number of zero															
8	C	90	<table><tr><th>Brønsted-Lowry Term</th><th>Definition</th></tr><tr><td>Acid</td><td>Proton donor to form conjugate base</td></tr><tr><td>Base</td><td>Proton acceptor to form conjugate acid</td></tr><tr><td>Conjugate Acid</td><td>Formed when Base accepts proton</td></tr><tr><td>Conjugate Base</td><td>Formed when Acid loses proton</td></tr></table>	Brønsted-Lowry Term	Definition	Acid	Proton donor to form conjugate base	Base	Proton acceptor to form conjugate acid	Conjugate Acid	Formed when Base accepts proton	Conjugate Base	Formed when Acid loses proton					
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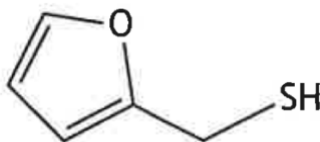
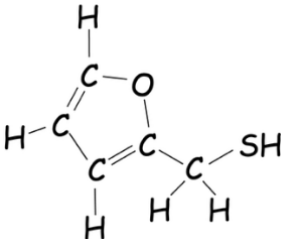
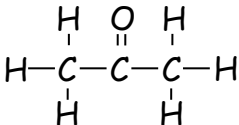
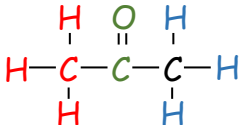
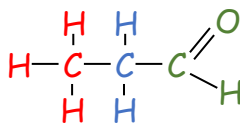
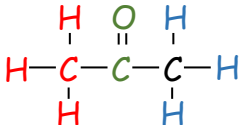
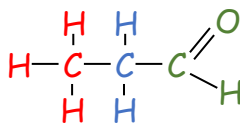
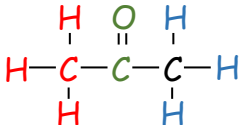
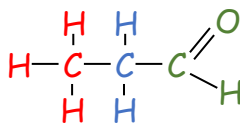
9	D	89	$\text{pH} = \text{pK}_a - \log_{10} \frac{[\text{Acid}]}{[\text{Salt}]}$ $\text{pH} = 4.83 - \log_{10} \frac{0.100}{0.200}$ $\text{pH} = 4.83 - \log_{10} (0.5)$ $\text{pH} = 4.83 - (-0.30)$ $\text{pH} = 5.13$																																																																				
10	A	47	☒ A NaCl is the salt formed by reaction of strong acid and strong alkali $\therefore$ pH of salt = 7 ☒ B Na ₂ CO ₃ is the salt formed by reaction of weak acid and strong alkali $\therefore$ pH of salt > 7 ☒ C Na ₂ SO ₃ is the salt formed by reaction of weak acid and strong alkali $\therefore$ pH of salt > 7 ☒ D CH ₃ COONa is the salt formed by reaction of weak acid and strong alkali $\therefore$ pH of salt > 7																																																																				
11	B	71	<table><tr><td>Property</td><td>0.1 mol l⁻¹ ethanoic acid</td><td>0.1 mol l⁻¹ hydrochloric acid</td></tr><tr><td>pH</td><td>Higher (partial dissociation = fewer H⁺ ions)</td><td>Lower (full dissociation = more H⁺ ions)</td></tr><tr><td>Conductivity</td><td>Lower (partial dissociation = fewer ions to conduct)</td><td>Higher (full dissociation = more ions to conduct)</td></tr><tr><td>Rate of Reaction with Magnesium</td><td>Slower (partial dissociation = fewer H⁺ ions to react)</td><td>Faster (full dissociation = more H⁺ ions to react)</td></tr></table>	Property	0.1 mol l ⁻¹ ethanoic acid	0.1 mol l ⁻¹ hydrochloric acid	pH	Higher (partial dissociation = fewer H ⁺ ions)	Lower (full dissociation = more H ⁺ ions)	Conductivity	Lower (partial dissociation = fewer ions to conduct)	Higher (full dissociation = more ions to conduct)	Rate of Reaction with Magnesium	Slower (partial dissociation = fewer H ⁺ ions to react)	Faster (full dissociation = more H ⁺ ions to react)																																																								
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12	B	80	Number of each reactant in the rate determining step gives the order for each reactant. • 2 molecules of NO in r.d.s. $\therefore$ order of reaction with respect to NO = 2 • 1 molecule of H ₂ in r.d.s. $\therefore$ order of reaction with respect to H ₂ = 1 Rate equation for reaction rate = k[NO] ² [H ₂] and overall order = 2+1 = 3																																																																				
13	D	79	The energy change for the formation of one mole of a substance from elements in their standard state ☒ A oxygen needs to be O ₂ molecules for the element to be in its standard state ☒ B none of the three reactants are in their standard state ☒ C calcium is a solid in its standard state ☒ D reactants are in their standard states and 1 mole of the substance is formed																																																																				
14	C	76	A reaction with a positive S° value is a reaction with an increase in disorder ☒ A Reaction has a decrease in disorder as two reactants join to become one product ☒ B Reaction has a decrease in disorder as two reactants join to become one liquid product ☒ C Reaction has an increase in disorder as a solid is removed and a gas is formed ☒ D Reaction has a decrease in disorder as a gas is removed and solid is formed																																																																				
15	C	66	<table><tr><td rowspan="5"></td><td colspan="2">Type of Bond</td><td>C - C</td><td>C = C</td><td>C - H</td><td>N = C</td><td>N - C</td></tr><tr><td colspan="2">Number of bonds</td><td>2</td><td>2</td><td>5</td><td>1</td><td>1</td></tr><tr><td colspan="2">Type of Bond</td><td>C - C</td><td>C = C</td><td>C - H</td><td>N = C</td><td>N - C</td></tr><tr><td>Sigma</td><td>σ</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Pi</td><td>π</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td colspan="2">Sigma bonds</td><td>=</td><td>2</td><td>2</td><td>5</td><td>1</td><td>1</td></tr><tr><td colspan="2"></td><td>=</td><td>C - C</td><td>+ C = C</td><td>+ C - H</td><td>+ N = C</td><td>+ N - C</td></tr><tr><td colspan="2"></td><td>=</td><td>2σ</td><td>+ 2σ</td><td>+ 5σ</td><td>+ 1σ</td><td>+ 1σ</td></tr><tr><td colspan="2"></td><td>=</td><td colspan="5">11σ</td></tr></table>		Type of Bond		C - C	C = C	C - H	N = C	N - C	Number of bonds		2	2	5	1	1	Type of Bond		C - C	C = C	C - H	N = C	N - C	Sigma	σ	1	1	1	1	1	Pi	π	0	1	0	1	0	Sigma bonds		=	2	2	5	1	1			=	C - C	+ C = C	+ C - H	+ N = C	+ N - C			=	2σ	+ 2σ	+ 5σ	+ 1σ	+ 1σ			=	11σ				
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16	B	74	☒ A Both can be made by nucleophilic substitution ☒ B pure ethers like ethoxyethane do not have hydrogen bonding between molecules ☒ C both compounds have the formula C ₄ H ₁₀ O and have the same gram formula mass ☒ D both compounds are carbon-based and are flammable																																																																				
17	A	48	<table><tr><th rowspan="2">Answer</th><th colspan="2">Type of Compound</th></tr><tr><th>Compound 1</th><th>Compound 2</th></tr><tr><td>A</td><td>tertiary haloalkane</td><td>tertiary amine</td></tr><tr><td>B</td><td>primary amine</td><td>tertiary haloalkane</td></tr><tr><td>C</td><td>primary haloalkane</td><td>tertiary amine</td></tr><tr><td>D</td><td>secondary amine</td><td>secondary haloalkane</td></tr></table>	Answer	Type of Compound		Compound 1	Compound 2	A	tertiary haloalkane	tertiary amine	B	primary amine	tertiary haloalkane	C	primary haloalkane	tertiary amine	D	secondary amine	secondary haloalkane																																																			
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18	C	70	☒ A addition reactions do not take place on benzene rings ☒ B nitrate NO_3^- ions are not involved in this electrophilic substitution reaction ☒ C NO_2^+ is the electrophile in this electrophilic substitution reaction ☒ D electrophiles are attracted to the delocalised electrons in benzene.
19	D	76	☒ A optical isomers (enantiomers) are mirror images of each other ☒ B geometric isomers can be based around $\text{C}=\text{C}$ double bonds and ring structures ☒ C there can be differences in physical & chemical properties due to the difference shapes ☒ D $\text{C}=\text{C}$ double bonds and ring structures can provide the restricted rotation needed for geometric isomerism where two different groups are attached across $\text{C}=\text{C}$ or ring
20	C	48	☒ A $n = v \times c = 0.2 \text{ litres} \times 0.12 \text{ mol l}^{-1} = 0.024 \text{ mol f.u.}$ but 3 Cl^- ions per f.u. $\therefore 0.072 \text{ mol -ve ions}$ ☒ B $n = v \times c = 0.3 \text{ litres} \times 0.15 \text{ mol l}^{-1} = 0.045 \text{ mol f.u.}$ but 1 I^- ions per f.u. $\therefore 0.045 \text{ mol -ve ions}$ ☒ C $n = v \times c = 0.4 \text{ litres} \times 0.10 \text{ mol l}^{-1} = 0.040 \text{ mol f.u.}$ but 2 NO_3^- ions per f.u. $\therefore$ 0.080 mol -ve ions ☒ D $n = v \times c = 0.5 \text{ litres} \times 0.10 \text{ mol l}^{-1} = 0.050 \text{ mol f.u.}$ but 1 SO_4^{2-} ions per f.u. $\therefore 0.050 \text{ mol -ve ions}$
21	A	74	☒ A radio waves are used to flip hydrogen nuclei from lower to higher energy alignment ☒ B ultraviolet is used in the initiation reaction in free radical chain reactions ☒ C Infrared is absorbed by specific bonds allowing the identification of type of bond ☒ D visible light is absorbed in d-d transitions in transition metals transmitting another colour
22	B	54	☒ A sodium hydroxide cannot be used as a primary standard as it absorbs moisture from air ☒ B sodium carbonate is a primary standard used to standardise acid like hydrochloric acid ☒ C potassium dichromate needs acid to become an oxidising agent. Not used to standardise acid ☒ D potassium hydrogen phthalate is used to standardise alkalis not acids
23	D	74	☒ A impurities lower melting point $\therefore$ purification should increase melting point ☒ B impurities lower melting point $\therefore$ purification should increase melting point ☒ C impurities widen the melting point $\therefore$ purification would narrow the range ☒ D purification should increase the melting point and narrow the range
24	B	42	☒ A carotene is a hydrocarbon so water cannot be used as the solvent for carotene ☒ B carotene is a hydrocarbon and is soluble in hexane and green-blue filter should be used ☒ C green-blue filter is used as it is complementary to the colour orange ☒ D green-blue filter is used as it is complementary to the colour orange
25	C	55	☒ A Step 1 does not remove soluble impurities as they are dissolved into the hot solvent ☒ B Hot filtration of resulting mixture in Step 2 to remove insoluble impurities ☒ C Hot filtration removes insoluble impurities. Filtering, washing and drying separates soluble impurities ☒ D Hot filtration of resulting mixture in Step 2 to remove insoluble impurities

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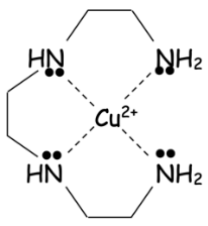
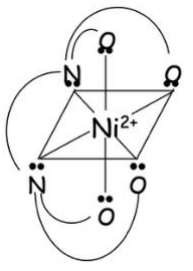
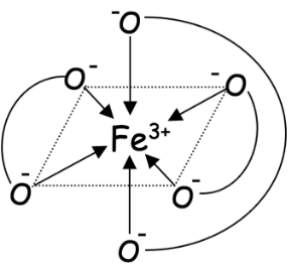
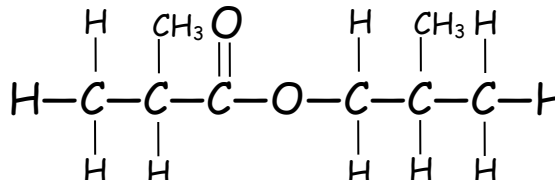
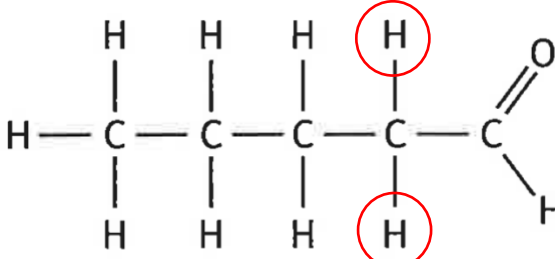
Long Qu	Answer	Reasoning										
1a	non-superimposable mirror images	Optical isomers are asymmetric non-superimposable mirror images formed when a chiral carbon has four different groups attached to it.										
1b		Mirror image diagrams  L-aspartic Acid D-aspartic Acid or "Swap Two Groups" Example  L-aspartic Acid D-aspartic Acid										
1c	One answer from:	rotates by same $\left\{ \begin{array}{c} \text{amount} \\ \text{angle} \end{array} \right\}$ in $\left\{ \begin{array}{c} \text{opposite} \\ \text{different} \end{array} \right\}$ directions										
1d	racemic	A racemic mixture has equal concentrations of both optical isomers and this makes the solution optically inactive giving no rotation of plane-polarised light.										
2a	Answer to include:	Side-on overlap of unhybridised p-orbitals (perpendicular to the σ -axis)										
2b(i)	One answer from:	<table><tr><td>the more hydroxyl/hydroxy/OH groups the stronger the acid</td><td>the longer the alkyl group/carbon chain the stronger the acid</td><td>ethyl phenols are stronger than methyl phenols</td></tr><tr><td colspan="2">if the alkyl/methyl/ethyl/carbon chain group is on carbon number 4 position the acid is stronger (than if it is on carbon number 2 position)</td><td>the presence of an alkyl/methyl/ethyl group lowers the strength.</td></tr></table>	the more hydroxyl/hydroxy/OH groups the stronger the acid	the longer the alkyl group/carbon chain the stronger the acid	ethyl phenols are stronger than methyl phenols	if the alkyl/methyl/ethyl/carbon chain group is on carbon number 4 position the acid is stronger (than if it is on carbon number 2 position)		the presence of an alkyl/methyl/ethyl group lowers the strength.				
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2b(ii)	2.77×10^{-6}	<table><tr><td>$\text{pH} = \frac{1}{2}\text{pK}_a - \frac{1}{2}\log_{10}c$</td><td>$\text{pH} = 5.56$</td></tr><tr><td>$\text{pH} = \frac{1}{2} \times (10.29) - \frac{1}{2}\log_{10}(0.150)$</td><td>$-\log_{10}[\text{H}^+] = 5.56$</td></tr><tr><td>$\text{pH} = 5.145 - \frac{1}{2} \times (-0.824)$</td><td>$\log_{10}[\text{H}^+] = -5.56$</td></tr><tr><td>$\text{pH} = 5.145 - (-0.412)$</td><td>$[\text{H}^+] = 10^{-5.56}$</td></tr><tr><td>$\text{pH} = 5.56$</td><td>$[\text{H}^+] = 2.77 \times 10^{-6} \text{ mol l}^{-1}$</td></tr></table>	$\text{pH} = \frac{1}{2}\text{pK}_a - \frac{1}{2}\log_{10}c$	$\text{pH} = 5.56$	$\text{pH} = \frac{1}{2} \times (10.29) - \frac{1}{2}\log_{10}(0.150)$	$-\log_{10}[\text{H}^+] = 5.56$	$\text{pH} = 5.145 - \frac{1}{2} \times (-0.824)$	$\log_{10}[\text{H}^+] = -5.56$	$\text{pH} = 5.145 - (-0.412)$	$[\text{H}^+] = 10^{-5.56}$	$\text{pH} = 5.56$	$[\text{H}^+] = 2.77 \times 10^{-6} \text{ mol l}^{-1}$
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2c(i)	Answer to include:	<table><tr><td>1st mark (one from:)</td><td>absorbs $\left\{ \begin{array}{c} \text{higher energy} \\ \text{shorter wavelength} \\ \text{higher frequency} \end{array} \right\}$</td><td>changes from absorbing yellow-green to green-blue or 560-580 nm to 480-490 nm</td><td>larger energy gap between HOMO and LUMO</td></tr><tr><td>2nd mark (one from:)</td><td>fewer atoms/electrons/bonds in the conjugated system</td><td>shorter/smaller conjugated system</td><td>conjugation decreases</td></tr></table>	1 st mark (one from:)	absorbs $\left\{ \begin{array}{c} \text{higher energy} \\ \text{shorter wavelength} \\ \text{higher frequency} \end{array} \right\}$	changes from absorbing yellow-green to green-blue or 560-580 nm to 480-490 nm	larger energy gap between HOMO and LUMO	2 nd mark (one from:)	fewer atoms/electrons/bonds in the conjugated system	shorter/smaller conjugated system	conjugation decreases		
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2 nd mark (one from:)	fewer atoms/electrons/bonds in the conjugated system	shorter/smaller conjugated system	conjugation decreases									
2c(ii)	4.1-6.1 or 5.1 $\pm$ 1	$\text{pH} = \text{pK}_{\text{In}} \pm 1$ $\text{pK}_{\text{In}} = -\log_{10}(7.9 \times 10^{-6})$ 5.1 $\text{pH range} = 5.1 \pm 1$ $= 4.1-6.1$										
3a	gravimetric	Gravimetric analysis is used to determine the mass of an element or compound in a substance. The substance is converted into another substance which can be readily isolated and purified. In this example, the chloride ions in seas water are removed by a precipitation reaction using silver (I) nitrate and collected by filtration and mass measured.										
3b	Add more silver(I) nitrate solution (to filtrate to see if there is any more precipitation)	Once the precipitate has been removed by filtration then the filtrate can be tests with a small addition of silver(I) nitrate solution to closely observe if there is further precipitation. - Any precipitation observed means that not all Cl^- ions have been turned reacted - No further precipitation means that all Cl^- ions have been precipitated out.										

3c	19.3	gfm AgCl = (1×107.9)+(1×35.5) = 107.9 + 35.5 = 143.4g no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{0.779}{143.4} = 5.43 \times 10^{-3} \text{ mol AgCl} = 5.43 \times 10^{-3} \text{ mol Cl}^-$ 20cm³ diluted seawater $\longleftrightarrow 5.43 \times 10^{-3} \text{ mol}$ 100 cm³ diluted seawater $\longleftrightarrow 5.43 \times 10^{-3} \text{ mol} \times \frac{100}{20} = 0.0271 \text{ mol}$ $\therefore$ 50cm³ seawater $\longleftrightarrow 0.0271 \text{ mol}$ 1000cm³ seawater $\longleftrightarrow 0.0271 \text{ mol} \times \frac{1000}{50} = 0.543 \text{ mol}$ mass = no. of mol x gfm = 0.543 x 35.5 = 19.3 g per litre														
4a(i)A																
4a(i)B		When the intermediate forms, the temporary increase in stability. This stability lowers the enthalpy a little and hence the dip at the top of the hill where the intermediate is formed.														
4a(ii)A	One answer from:	entropy of surroundings is increasing	total entropy (of system and surroundings) is increasing	reaction is exothermic or ΔH is negative	heat energy is released											
4a(ii)B	-80	ΔH = H _{products} - H _{reactants} = -60 - 20 = -80 kJ mol ⁻¹														
4a(ii)C	606 or 610	ΔG° = ΔH° - TΔS° = 0 $\therefore T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{-80 \times 1000 \text{ J mol}^{-1}}{-132 \text{ J K}^{-1} \text{ mol}^{-1}} = 606 \text{ K}$														
4b(i)	more stable/stabilisation or the intermediate for the minor product is less stable	H from H-Br more likely to attach to end carbon as there is more space for the H-Br molecule to collide with the end carbon. This results in 2-bromopropane being the major product in accordance with Markovnikov's Rule. H from H-Br less likely to attach to middle carbon as there is less space for the H-Br molecule to collide with the middle carbon. This results in 1-bromopropane being the minor product in accordance with Markovnikov's Rule.														
4b(ii)	Open Ended Question: <table><tr><th colspan="4">Marks Awarded</th></tr><tr><td>3 marks</td><td>2 marks</td><td>1 mark</td><td>0 marks</td></tr><tr><td>3%</td><td>13%</td><td>35%</td><td>49%</td></tr></table>	Marks Awarded				3 marks	2 marks	1 mark	0 marks	3%	13%	35%	49%	3 mark answer Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	2 mark answer Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	1 mark answer Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
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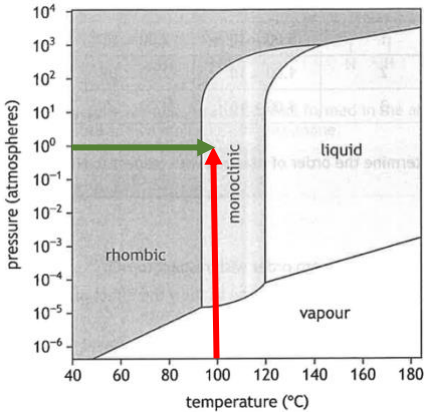
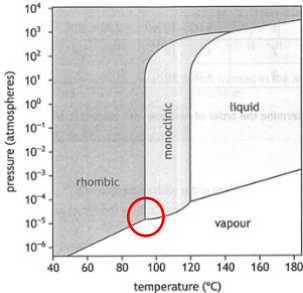
5a(i)	C ₅ H ₆ OS	  C₅H₆OS (any order of elements) Skeletal structure Full Structural Formula Molecular Formula																					
5a(ii)	[SH] ⁺ or ⁺ SH	The thiol group is a sulfur bonded to a hydrogen atom. Sulphur has r.a.m.=32 and hydrogen has r.a.m.=1 ∴ [SH] ⁺ fragment has m/z = 33																					
5b(i)	C=O or carbonyl	The infrared adsorption at 1710cm ⁻¹ is due to the stretching of an C=O bond in an aldehyde or ketone. (The only other infrared stretch at 1710cm ⁻¹ is the C=O bond in a carboxylic acid but it can be ruled out because carboxylic acids contain 2 oxygen atoms per molecule but the formula of the molecule is C ₃ H ₆ O)																					
5b(ii)A		There is only one peak in the Proton NMR spectrum at chemical shift = 2.2. This corresponds to aldehyde or ketone CH ₃ C=O.																					
5b(ii)B	3	<table><tr><td>Propanone (Compound A)</td><td>Propanal (Compound B)</td></tr><tr><td></td><td></td></tr><tr><td>There should be 1 vertical line for propanone corresponding to</td><td>There should be 3 vertical lines for propanal corresponding to</td></tr><tr><td><table><tr><td>CH₃-C=O</td><td>O=C-CH₃</td></tr><tr><td>2.7-2.0</td><td>2.7-2.0</td></tr></table></td><td><table><tr><td>CH₃-R</td><td>R-CH₂-CHO</td><td>-CH₂-CHO</td></tr><tr><td>1.5-0.9</td><td></td><td>2.7-2.0</td></tr></table></td></tr><tr><td colspan="3">There is only one vertical line as the two side are identical.</td></tr></table>	Propanone (Compound A)	Propanal (Compound B)			There should be 1 vertical line for propanone corresponding to	There should be 3 vertical lines for propanal corresponding to	<table><tr><td>CH₃-C=O</td><td>O=C-CH₃</td></tr><tr><td>2.7-2.0</td><td>2.7-2.0</td></tr></table>	CH ₃ -C=O	O=C-CH ₃	2.7-2.0	2.7-2.0	<table><tr><td>CH₃-R</td><td>R-CH₂-CHO</td><td>-CH₂-CHO</td></tr><tr><td>1.5-0.9</td><td></td><td>2.7-2.0</td></tr></table>	CH ₃ -R	R-CH ₂ -CHO	-CH ₂ -CHO	1.5-0.9		2.7-2.0	There is only one vertical line as the two side are identical.		
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5d(ii)	9.71	Mass of caffeine in aqueous layer = 0.150 - 0.136 = 0.014g $K = \frac{[\text{caffeine}]_{\text{dichloromethane}}}{[\text{caffeine}]_{\text{water}}} = \frac{0.136}{0.014} = 9.71$																					

5d(iii)		● The added black dot represents the caffeine in the impure sample (S) <u>On Diagram:</u> There must be a dot to represent the caffeine present at the same distance travelled by the pure caffeine (C) ○ The white dot represents impurities. The number of impurity dots will depend on the number of impurities in the sample. <u>On Diagram:</u> The distance travelled by the impurity dot(s) depends on the polarity and size of the impurities. At least one dot must be drawn either vertically above or below the caffeine dot. Multiple dots are also correct				
6a(i)	1 mark for: 25cm³ or 0.025 litres 1 mark for: use of pipette/burette <u>and</u> making up to the mark/line (with deionised water)	no. of mol = volume × concentration = 0.5 litres × 0.01 mol l⁻¹ = 0.005 mol volume = $\frac{\text{no. of mol}}{\text{concentration}}$ = $\frac{0.005 \text{ mol}}{0.2 \text{ mol l}^{-1}}$ = 0.025 litres = 25cm³ <table><tr><td>25cm³ of 0.2 mol l⁻¹ KMnO₄ solution pipetted into 500cm³ volumetric flask</td><td>Flask filled up to near mark with deionised water and a dropper used to add few drops up to mark on flask.</td><td>Mix thoroughly before use.</td></tr></table>	25cm ³ of 0.2 mol l ⁻¹ KMnO ₄ solution pipetted into 500cm ³ volumetric flask	Flask filled up to near mark with deionised water and a dropper used to add few drops up to mark on flask.	Mix thoroughly before use.	
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6a(ii)	65.8	no. of mol = volume × concentration = 0.0118 litres × 0.010 mol l⁻¹ = 0.000118 mol $\begin{array}{ccccccc} \text{MnO}_4^- & + & 8\text{H}^+ & + & 5\text{Fe}^{2+} & \longrightarrow & \text{Mn}^{2+} + 4\text{H}_2\text{O} + 5\text{Fe}^{3+} \\ 1\text{mol} & & & & 5\text{mol} & & \\ 0.000118 \text{ mol} & & & & 0.00059 \text{ mol} & & \end{array}$ mass = no. of mol × gfm = 0.00059 × 55.8 = 0.0329g 25cm³ sample of solution contains 0.0329g 250cm³ flask contains 0.0329g × ²⁵⁰/₂₅ = 0.329g 5 tablets contains 0.329g 1 tablet contains 0.329g × ¹/₅ = 0.0658g = 65.8mg				
6a(iii)	acidified permanganate would oxidise Cl ⁻ ions to Cl ₂	Acidified permanganate solution is a strong enough oxidising agent to cause the oxidation of Cl⁻ ions into Cl₂ gas $2\text{MnO}_4^- + 16\text{H}^+ + 10\text{Cl}^- \longrightarrow 2\text{Mn}^{2+} + 8\text{H}_2\text{O} + 5\text{Cl}_2$				
6b(i)	Answer to include:	<table><tr><td>1st mark</td><td> purple 380-400nm complementary colour to green } is absorbed</td></tr><tr><td>2nd mark</td><td>Electrons move to higher energy d orbitals</td></tr></table>	1 st mark	purple 380-400nm complementary colour to green } is absorbed	2 nd mark	Electrons move to higher energy d orbitals
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6b(ii)	One answer from both sections required for 1 mark:	one answer from: the water ligand water in complex OH₂ } { donates a proton/H⁺ acts as an acid <u>and</u> one answer from: the water molecule water solvent H₂O } { accepts a proton/H⁺ acts as a base				
7a	CO ₂ or carbon dioxide	<chem>C9H11O4N</chem> → <chem>C8H11O2N</chem> + CO₂				

7b(i)	1.03×10^{-4} or 0.000103	$\text{mass} = 125\text{mg} = 0.125\text{g}$ $\text{gfm} = 197\text{g}$ $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{0.125}{197} = 6.35 \times 10^{-4} \text{ mol}$ 100% Yield = $6.35 \times 10^{-4} \text{ mol}$ 70.5% Yield = $6.35 \times 10^{-4} \text{ mol} \times \frac{70.5}{100}$ = $4.47 \times 10^{-4} \text{ mol}$ $\text{concentration} = \frac{\text{no. of mol}}{\text{volume}} = \frac{4.47 \times 10^{-4} \text{ mol}}{4.35 \text{ litres}} = 1.03 \times 10^{-4} \text{ mol l}^{-1}$																				
7b(ii)A	(enzyme) inhibitor	<table><tr><th>Classification</th><th>Definition</th></tr><tr><td>Agonist</td><td>Mimics natural compound and binds to receptors to produce response</td></tr><tr><td>Antagonist</td><td>Prevents the natural compound from binding to receptor blocking response</td></tr><tr><td>Inhibitor</td><td>Binds to enzyme active site preventing the. reaction catalysed by enzyme</td></tr></table>			Classification	Definition	Agonist	Mimics natural compound and binds to receptors to produce response	Antagonist	Prevents the natural compound from binding to receptor blocking response	Inhibitor	Binds to enzyme active site preventing the. reaction catalysed by enzyme										
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7b(ii)B	6	Three groups in compounds can exhibit hydrogen bonding • 3 from 3x-OH • 1 from -NH- • 2 from -NH₂																				
8a	254	$E = \frac{L \times h \times c}{\lambda} = \frac{6.02 \times 10^{23} \text{ mol}^{-1} \times 6.63 \times 10^{-34} \text{ J s} \times 3 \times 10^8 \text{ m s}^{-1}}{471 \times 10^{-9} \text{ m}}$ $= 254220 \text{ J mol}^{-1}$ $= 254 \text{ kJ mol}^{-1}$																				
8b(i)		Homolytic fission happens when a pair of electrons split evenly and one electron goes to each end of the bond creating two free radicals. Curly arrows with a single head (shown here) represent the movement on one electron only.																				
8b(ii)	One answer from:	<table><tr><td>(Complex) mixture of products (formed)</td><td>Many products (formed)</td><td>Side reactions</td></tr></table>			(Complex) mixture of products (formed)	Many products (formed)	Side reactions															
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10a(i)	904cm ³	$\text{mass} = 1.00\text{g}$ $\text{gfm} = 197\text{g}$ $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{1.00}{197} = 0.00508 \text{ mol}$ $2\text{Au} + 4\text{NaCN} + \frac{1}{2}\text{O}_2 + \text{H}_2\text{O} + 2\text{OH}^- \longrightarrow 2[\text{Au}(\text{CN})_2]^- + 4\text{NaOH}$ 2mol 4mol 0.00508mol 0.0102mol $\text{gfm NaCN} = (1 \times 23) + (1 \times 12) + (1 \times 14) = 23 + 12 + 14 = 49\text{g}$ $\text{mass} = \text{no. of mol} \times \text{gfm} = 0.0102 \times 49 = 0.497\text{g}$ $0.0550\% = 0.00550\text{g per } 100\text{cm}^3$ 0.005500g in 100cm³ 0.497g in 100cm³ $\times \frac{0.497}{0.00550} = 904\text{cm}^3$																				

10a(ii)A	total number of bonds from ligands to central metal ion or number of bonds the metal ion forms or number of pairs of electrons donated/shared to the metal ion	Examples of ligands forming complexes with central transition metal ions:																											
																													
		coordination number = 4	coordination number = 6	coordination number = 6																									
10a(ii)B	dicyanidoaurate(I)	dicyanidoaurate(I) = [Au(CN)₂]⁻ no. of ligands CN⁻ ligand metal name negative charge on complex metal ion <table><tr><td colspan="2">Neutral ligands include:</td><td colspan="2">Negative Ligands include:</td><td>Central Ion:</td><td>Charge:</td></tr><tr><td>Ligand</td><td>Name</td><td>Ligand</td><td>Name</td><td rowspan="4">Positive Complex: metals keep their name Negative Complex: Metals end in ATE e.g. Cuprate, Ferrate, Cobaltate</td><td rowspan="4">Charge of central ion is converted into roman numerals and put in brackets</td></tr><tr><td>OH₂</td><td>aqua</td><td>Chloride Cl⁻</td><td>chlorido</td></tr><tr><td>NH₃</td><td>ammine</td><td>Cyanide CN⁻</td><td>cyanido</td></tr><tr><td>CO</td><td>carbonyl</td><td>Nitrite NO₂⁻</td><td>nitrito</td></tr></table> NB: Multiple ligands are listed in alphabetical order ∴ ammine listed before chloride				Neutral ligands include:		Negative Ligands include:		Central Ion:	Charge:	Ligand	Name	Ligand	Name	Positive Complex: metals keep their name Negative Complex: Metals end in ATE e.g. Cuprate, Ferrate, Cobaltate	Charge of central ion is converted into roman numerals and put in brackets	OH ₂	aqua	Chloride Cl ⁻	chlorido	NH ₃	ammine	Cyanide CN ⁻	cyanido	CO	carbonyl	Nitrite NO ₂ ⁻	nitrito
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10b(i)	reduction	Reduction: Au ⁺ + e ⁻ → Au																											
10b(ii)	Answer to include:	<table><tr><td>filter</td><td>wash</td><td>dry/heat</td><td>to constant mass</td></tr></table>				filter	wash	dry/heat	to constant mass																				
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10b(iii)	50.8	21.5ppm = 21.5mg per 1kg rock 91.5% yield = 1kg rock 100% yield = 1kg rock x ¹⁰⁰/_{91.5} = 1.09kg rock 21.5mg found in 1.09kg rock 0.0215g found in 1.09kg rock 1.00g found in 1.09kg rock x ^{1.00}/_{0.0215} = 50.8kg rock																											
11a(i)	Lithium Aluminium Hydride or LiAlH ₄	Lithium Aluminium Hydride is a reducing agent which would cause an aldehyde (methylpropanal) to be reduced to a primary alcohol (methylpropan-1-ol)																											
11a(ii)	Oxidation	Acidified dichromate solution containing Cr ₂ O ₇ ²⁻ ions is an oxidising agent. This oxidising agent would oxidise an aldehyde (methylpropanal) to a carboxylic acid (methylpropanoic acid)																											
11a(iii)	Diagram showing:																												
11b(i)A	One circle drawn from:																												

11b(i)B	Diagram of 2,2-dimethylpropanoic acid																
11b(ii)A		Examine closely original diagram to spot relationships between reactants and products Make the changes to the reactants specified in the question and follow original															
11b(ii)B PART I	Two different alpha hydrogen atoms/alpha hydrogen atoms present on different positions. or Butanone is asymmetrical. or Two different intermediates formed. or Intermediates formed have different stabilities.	Carbon C₃ on left butanone molecule reacts with C₂ of right butanone molecule: Product formed: 4-hydroxy-3,4-dimethylhexan-2-one Carbon C₁ on left butanone molecule reacts with C₂ of right butanone molecule: Product formed: (Compound Z) 5-hydroxy-5-methylheptan-3-one															
11b(ii)B PART II	5-hydroxy-5-methyl heptan-3-one	5-hydroxy-5-methylheptan-3-one -OH hydroxyl group on C₅ -CH₃ methyl group on C₅ 7 carbons on main chain C=O carbonyl group on C₃															
12a(i)	1 st order	<table><tr><th>Experiment</th><th>[Na₂S₂O₃]</th><th>[HCl]</th><th>Effect on Rate</th><th>Order of Reactant</th></tr><tr><td>1+2</td><td>x2</td><td>no change</td><td>x2</td><td>[Na₂S₂O₃]¹</td></tr><tr><td>2+3</td><td>no change</td><td>x2</td><td>no change</td><td>[HCl]⁰</td></tr></table>	Experiment	[Na ₂ S ₂ O ₃]	[HCl]	Effect on Rate	Order of Reactant	1+2	x2	no change	x2	[Na ₂ S ₂ O ₃] ¹	2+3	no change	x2	no change	[HCl] ⁰
Experiment	[Na ₂ S ₂ O ₃]	[HCl]	Effect on Rate	Order of Reactant													
1+2	x2	no change	x2	[Na ₂ S ₂ O ₃] ¹													
2+3	no change	x2	no change	[HCl] ⁰													
12a(ii)		Zero Order Kinetics: rate of reaction is same at all concentrations of HCl															
12a(iii)	Rate = k[Na ₂ S ₂ O ₃]	rate = k × [Na ₂ S ₂ O ₃] ¹ × [HCl] ⁰ = k[Na ₂ S ₂ O ₃]															

12a(iv)	0.131 s^{-1}	Using Experiment 1 (although experiments 2+3 can also be used) $k = \frac{\text{rate}}{[\text{Na}_2\text{S}_2\text{O}_3]} = \frac{6.55 \times 10^{-3} \text{ mol l}^{-1} \text{ s}^{-1}}{5.00 \times 10^{-2} \text{ mol l}^{-1}} = 0.131 \text{ s}^{-1}$
12b(i)	monoclinic	
12b(ii)A		There is only one point on the graph where rhombic, monoclinic and vapour intersect. These conditions (~92°C and 10^{-5} atmospheres) are the only point where all three are in equilibrium with each other.
12b(ii)B	Rhombic, liquid and vapour	There is no junction of lines on the graph where rhombic, liquid and vapour meet

2025 Advanced Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	A	93	1	2	4	0	0
2	D	7	22	4	67	0	0
3	D	13	12	14	61	0	0
4	B	10	53	21	16	0	0
5	A	42	16	17	25	1	0
6	C	19	30	39	12	1	0
7	A	48	36	8	7	1	0
8	C	5	1	90	4	0	0
9	D	2	4	4	89	1	0
10	A	47	10	21	22	1	0
11	B	7	71	7	15	0	0
12	B	15	80	4	1	0	0
13	D	8	2	10	79	1	0
14	C	10	4	76	9	1	0
15	C	6	25	66	2	0	0
16	B	11	74	8	6	0	0
17	A	48	44	6	1	0	0
18	C	6	14	70	10	0	0
19	D	6	12	5	76	0	0
20	C	12	10	48	30	1	0
21	A	74	10	12	3	0	0
22	B	15	54	20	10	1	0
23	D	3	20	3	74	0	0
24	B	51	42	3	3	1	0
25	C	19	5	55	21	0	0

2025 Advanced Higher Chemistry Section 2 Marks Data Analysis

2025 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1a	1	0.6	56.7%	44	1.1%
Q1b	1	0.6	57.4%	27	4.4%
Q1c	1	0.2	23.7%	38	6.6%
Q1d	1	0.8	81.1%	44	4.7%
Q2a	1	0.7	65.1%	35	2.2%
Q2bi	1	0.4	42.7%	46	1.6%
Q2bii	3	2.4	78.8%	51	2.4%
Q2ci	2	0.6	28.0%	45	5.0%
Q2cii	1	0.6	55.7%	45	5.2%
Q3a	1	0.6	60.5%	38	3.5%
Q3b	1	0.7	73.2%	40	2.5%
Q3c	2	0.7	37.2%	48	3.6%
Q4aiA	1	0.4	40.0%	47	2.7%
Q4aiB	1	0.6	55.3%	16	0.6%
Q4aiiA	1	0.4	42.9%	42	10.4%
Q4aiiB	1	0.7	66.3%	26	0.6%
Q4aiiC	2	1.6	78.7%	47	2.0%
Q4bi	1	0.5	49.6%	51	6.4%
Q4bii	3	0.7	23.4%	54	16.5%
Q5ai	1	0.8	76.9%	40	0.4%
Q5aii	1	0.3	34.4%	44	3.7%
Q5bi	1	0.9	93.2%	32	0.7%
Q5biiA	1	0.6	62.1%	45	2.4%
Q5biiB	1	0.4	44.9%	40	2.9%
Q5ci	2	1.1	55.2%	53	3.8%
Q5cii	1	0.5	46.0%	47	12.0%
Q5di	2	0.6	29.7%	52	8.8%
Q5dii	2	0.8	41.7%	50	4.8%
Q5diii	1	0.6	59.2%	41	1.5%
Q6ai	2	1.1	52.8%	49	5.0%
Q6aii	3	2.1	70.1%	56	3.3%
Q6aiii	1	0	2.8%	17	9.5%
Q6bi	2	0.9	43.1%	52	4.1%
Q6bii	1	0.2	15.2%	32	4.4%
Q7a	1	0.3	30.6%	7	1.0%
Q7bi	2	1.4	68.9%	44	2.6%
Q7biiA	1	0.5	50.1%	35	1.8%
Q7biiB	1	0.4	35.1%	26	1.6%
Q8a	2	1.8	87.9%	35	0.7%
Q8bi	1	0.5	50.7%	36	21.9%
Q8bii	1	0.3	28.4%	34	5.2%
Q9	3	1.3	44.6%	46	7.0%
Q10ai	2	0.7	35.1%	48	13.3%
Q10aiiA	1	0.5	50.2%	35	2.5%
Q10aiiB	1	0.5	49.3%	37	1.4%
Q10bi	1	0.1	15.0%	29	5.4%
Q10bii	2	0.7	35.6%	37	8.8%
Q10biii	2	1	52.0%	58	7.4%
Q11ai	1	0.6	63.1%	52	2.7%
Q11aii	1	0.7	68.6%	49	3.4%
Q11aiii	1	0.3	33.4%	52	16.5%
Q11biA	1	0.8	81.7%	30	3.0%
Q11biB	1	0.5	51.2%	37	3.7%
Q11biiA	1	0.2	18.1%	39	4.7%
Q11biiBI	1	0.1	10.0%	27	11.9%
Q11biiBII	1	0.6	59.9%	26	3.8%
Q12ai	1	0.9	89.3%	41	1.1%
Q12aii	1	0.6	64.7%	24	3.2%
Q12aiii	1	0.7	67.9%	42	2.9%
Q12aiv	2	1.5	74.9%	57	4.4%
Q12bi	1	1	95.6%	20	1.8%
Q12biiA	1	0.9	86.1%	20	1.9%
Q12biiB	1	0.9	86.1%	23	3.5%
Paper	85	44.8	52.7%		

2025 Advanced Higher Chemistry Section 2 Frequency of Marks Analysis

2025 Item	0	1	2	3
Q1a	43	57		
Q1b	43	57		
Q1c	76	24		
Q1d	19	81		
Q2a	35	65		
Q2bi	57	43		
Q2bii	11	5	20	64
Q2ci	63	18	19	
Q2cii	44	56		
Q3a	39	61		
Q3b	27	73		
Q3c	39	48	13	
Q4aiA	60	40		
Q4aiB	45	55		
Q4aiiA	57	43		
Q4aiiB	34	66		
Q4aiiC	10	23	67	
Q4bi	50	50		
Q4bii	49	35	13	3
Q5ai	23	77		
Q5aii	66	34		
Q5bi	7	93		
Q5biiA	38	62		
Q5biiB	55	45		
Q5ci	26	37	37	
Q5cii	54	46		
Q5di	55	31	14	
Q5dii	42	33	25	
Q5diii	41	59		
Q6ai	31	31	37	
Q6aii	8	7	52	33
Q6aiii	97	3		
Q6bi	37	39	23	
Q6bii	85	15		
Q7a	69	31		
Q7bi	8	47	46	
Q7biiA	50	50		
Q7biiB	65	35		
Q8a	2	20	78	
Q8bi	49	51		
Q8bii	72	28		
Q9	18	40	32	10
Q10ai	33	63	3	
Q10aiiA	50	50		
Q10aiiB	51	49		
Q10bi	85	15		
Q10bii	32	65	3	
Q10biii	28	40	32	
Q11ai	37	63		
Q11aii	31	69		
Q11aiii	67	33		
Q11biA	18	82		
Q11biB	49	51		
Q11biiA	82	18		
Q11biiB	90	10		
Q11biiC	40	60		
Q12ai	11	89		
Q12aii	35	65		
Q12aiii	32	68		
Q12aiv	14	23	63	
Q12bi	4	96		
Q12biiA	14	86		
Q12biiB	14	86		

2025 Advanced Higher Chemistry Section 3 Project Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.8	81.8%	30
2	3	1.9	63.9%	42
3a	2	1.4	68.0%	40
3b	1	0.4	43.7%	14
3c	1	0.8	82.4%	25
3d	1	0.8	83.2%	20
3e	1	0.6	62.3%	24
3f	1	0.7	68.4%	20
3g	1	0.6	61.7%	24
3h	1	0.6	63.4%	28
4a	4	2.7	67.0%	50
4b	1	0.8	84.0%	13
5	1	0.7	74.1%	33
6	1	0.8	83.1%	29
7	4	2.4	58.8%	48
8	1	0.8	84.9%	21
Paper	25	17	68.1%	

Item	Max Mark	0	1	2	3	4
1	1	18	82			
2	3	1	30	43	25	
3a	2	13	38	49		
3b	1	56	44			
3c	1	18	82			
3d	1	17	83			
3e	1	38	62			
3f	1	32	68			
3g	1	38	62			
3h	1	37	63			
4a	4	2	11	33	27	28
4b	1	16	84			
5	1	26	74			
6	1	17	83			
7	4	8	18	27	26	21
8	1	15	85			
Paper	25					

Item	Max Mark	Item Description
1	1	Brief abstract (summary) stating the overall aim and conclusion
2	3	Underlying chemistry
3a	2	Procedure
3b	1	Safety measures
3c	1	One procedure & second procedure/modification/control/ standardisation
3d	1	Experimental results providing evidence of duplicates
3e	1	Description of the correct use of the appropriate apparatus, chemicals,etc
3f	1	All relevant raw data is recorded
3g	1	Numerical data is appropriately presented
3h	1	Citations and references for three sources of internet/literature data
4a	4	Analysis of data with values calculated correctly using a chemical relationship , scatter, line or bar graph, chromatograms and spectra
4b	1	Calculated values stated to an appropriate number of significant figures
5	1	A valid conclusion that relates to the aim and is supported by data
6	1	A valid comparison of the experimental data with data from the internet/literature sources, or a comparison of duplicate experimental data
7	4	Evaluation including evaluation of data from the internet/literature, evaluative statements supported by justification, quantitative treatment of uncertainties
8	1	A clear and concise report with title, contents page and page numbers